

(ADAM) CHEOL WOO KIM

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ACADEMIC APPOINTMENTS

Harvard University

Postdoctoral Fellow in Computer Science

- Advisor: Milind Tambe

Boston, MA

Sep 2024 - Present

EDUCATION

Massachusetts Institute of Technology

PhD in Operations Research ; GPA: 5.0/5.0

- Advisor: Dimitris Bertsimas
- Thesis: *Predictive and Prescriptive Trees for Optimization and Control Problems*

Cambridge, MA

Aug 2024

Seoul National University School of Law

J.D. Candidate (withdrew to pursue PhD at MIT)

Seoul, Republic of Korea

Feb 2019 - Jul 2019

Seoul National University

B.A. in Economics (Summa Cum Laude); Major GPA: 4.24/4.30

- Concentration in Mathematical Economics and Microeconomic Theory

Seoul, Republic of Korea

Feb 2019

RESEARCH INTERESTS

AI Alignment, Reinforcement Learning with Human Feedback, Human-in-the-Loop Systems, ML-accelerated Optimization (Learning for Optimization), Generative AI for Decision-making

PAPERS

*: alphabetical or co-first author

Conference and Workshop Papers

1. *Preference Robustness for DPO with Applications to Public Health*
Cheol Woo Kim*, Shresth Verma*, Mauricio Tec*, Milind Tambe
AAAI Conference on Artificial Intelligence (AAAI), 2026.
2. *Lightweight Robust Direct Preference Optimization*
Cheol Woo Kim*, Shresth Verma*, Mauricio Tec*, Milind Tambe
NeurIPS Workshop: Reliable ML from Unreliable Data, 2025.
3. *Navigating the Social Welfare Frontier: Portfolios for Multi-objective Reinforcement Learning*
Cheol Woo Kim*, Jai Moondra*, Shresth Verma, Madeleine Pollack, Lingkai Kong, Milind Tambe, Swati Gupta
International Conference on Machine Learning (ICML), 2025.
4. *Robust Optimization with Diffusion Models for Green Security*
Lingkai Kong, Haichuan Wang, Yuqi Pan, **Cheol Woo Kim**, Mingxiao Song, Alayna Nguyen, Tonghan Wang, Haifeng Xu, Milind Tambe
Uncertainty in Artificial Intelligence (UAI), 2025.
5. *LLM-based Agent Simulation for Maternal Health Interventions: Uncertainty Estimation and Decision-focused Evaluation*
Sarah Martinson, Lingkai Kong, **Cheol Woo Kim**, Aparna Taneja, Milind Tambe
AAMAS Workshop on Autonomous Agents for Social Good, 2025.

Journal Papers (Including Under Revision)

1. *Optimal Control of Multiclass Fluid Queueing Networks: A Machine Learning Approach*
Dimitris Bertsimas*, **Cheol Woo Kim***
Operations Research (Minor Revision), 2024.
2. *Optimal Control of Fluid Restless Multi-armed Bandits: A Machine Learning Approach*
Dimitris Bertsimas*, **Cheol Woo Kim***, José Niño-Mora*
Machine Learning (Major Revision), 2024.
3. *A Machine Learning Approach to Two-stage Adaptive Robust Optimization*
Dimitris Bertsimas*, **Cheol Woo Kim***
European Journal of Operational Research, 2024.
4. *A Prescriptive Machine Learning Approach to Mixed Integer Convex Optimization*
Dimitris Bertsimas*, **Cheol Woo Kim***
INFORMS Journal on Computing, 2023.

PROFESSIONAL EXPERIENCE

Massachusetts Institute of Technology

Cambridge, MA

PhD Candidate

Sep 2019 - Aug 2024

Core Thesis Topic

- Learning-accelerated algorithms for optimization and control.

Application-driven Projects

- Developed machine learning models to predict patient flow at Beth Israel Deaconess Medical Center. The prediction models are fully integrated into their system.
- Collaborated with the OCP Group to develop a machine learning model to predict agricultural yield in Morocco combining image data and chemical composition data.
- Participated in a collaborative work to combat the spread of the Covid-19 pandemic:
(<https://www.covidanalytics.io/>)

Microsoft Research

Redmond, WA

Research Intern - Machine Learning and Optimization Group

May 2023 - Aug 2023

- Research Topic: Theory and algorithms for combinatorial optimization under uncertainty.
- Mentors: Ishai Menache, Marco Molinaro, Konstantina Mellou

Permanent Mission of the Republic of Korea to the United Nations

New York, NY

Research Intern - UN General Assembly Group

Jan 2014 - Apr 2014

TEACHING EXPERIENCE

Developed course materials, led recitations, and held office hours for the following courses:

TA, The Analytics Edge (MIT 15.071)

Spring 2024

- MBA course on statistics, machine learning, optimization, and data science.

TA, Optimization Methods (MIT 6.7200J/15.093J)

Fall 2022

- Graduate course on linear, nonlinear, combinatorial, robust, and semi-definite optimization.

TA, Introduction to Mathematical Programming (MIT 6.251J/15.081J)

Fall 2021

- PhD-level course on linear, nonlinear, discrete, combinatorial, robust, and semi-definite optimization.

TA, Robust Modeling, Optimization, and Computation (MIT 1.142J/15.094J) *Spring 2021*

- Graduate course on theory and applications of robust optimization.

REVIEWER SERVICES

Journal: Journal of Machine Learning Research, INFORMS Journal on Computing, INFORMS Journal on Optimization, European Journal of Operational Research, Transportation Science

Conference: AAAI 2026

AWARDS AND HONORS

Kwanjeong Educational Fellowship

Sep 2019 – Feb 2025

- Merit-based PhD scholarship.

Eminence Scholarship

Aug 2017

Department of Economics, Seoul National University

Seoul, Republic of Korea

- Awarded to top 1% in the economics department.

SKILLS

- **Programming:** Python (PyTorch, Scikit-learn), Julia (JuMP, Gurobi, Mosek), R
- **Languages:** Korean (Native), English (Fluent)